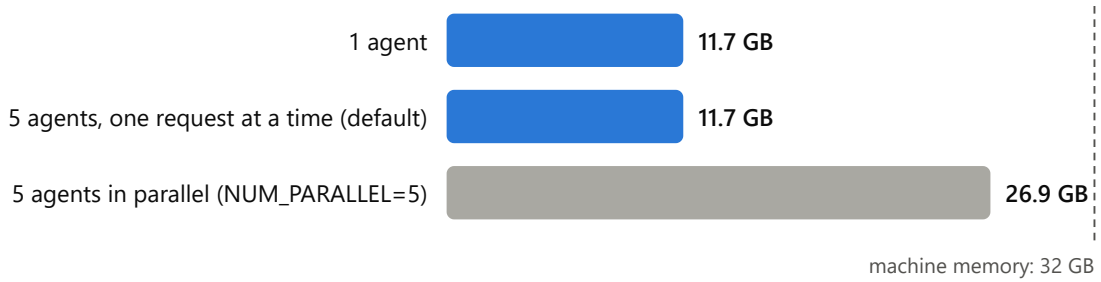
Does a model use the same memory with five agents as with one?

Measured with WhiteRabbitNeo 13B (Q4_K_M) on Apple Mac mini, Apple M6, 32 GB unified memory, Ollama 0.40.5.

> **Short answer.** It depends on the Ollama setting. With the default (one request at a time) five agents that share a model use **the same 11.7 GB** as one agent, but they are served one after another. If you let Ollama serve the five agents in parallel, memory grows to **26.9 GB** (2.3x as much) and the five answers arrive about 2.9x as fast.



What was measured

Setting	Memory	Time for 5 answers of 120 tokens	Total speed	Speed per agent
One request at a time (default)	11.7 GB	32.3 s	19 tok/s	18.4 tok/s while it is its turn
Parallel (5 at once)	26.9 GB	11.1 s	54 tok/s	10.8 tok/s each

Why it works this way

- **The model weights are loaded once** (7.9 GB here) and shared by every agent that uses the same model. More agents do not copy the model.
- **Each request slot needs its own context memory** (the "KV cache"): about 3.8 GB per slot at a context of 8,192 tokens. One slot gives 7.9 + 3.8 = 11.7 GB; 5 slots give 7.9 + 5 x 3.8 = 26.9 GB. Ollama reserves the slots when the model loads, whether or not they are busy.
- **With one slot the agents queue.** Memory stays low, but the fifth agent waits for the four before it.
- **Different models add up.** Five agents on five different models keep five models in memory (up to the limit set in Ollama; here 2 loaded at a time).

What to choose

- **Agents that mostly wait on each other or on tools:** keep one slot. Cheapest in memory.
- **Agents that all answer at the same moment:** use two or three slots as a compromise, or a smaller model. A 32 GB machine has only about 5 GB left over at 26.9 GB.
- **Shrink the cost per slot** with a smaller context window, or pick a model with memory-efficient attention (newer 7B and 8B models need far less context memory than Llama 2 13B).

One measurement per setting, on one machine and one model, with flash attention on and an 8-bit KV cache. Numbers for other models and context lengths will differ. The parallel run used a temporary second Ollama instance with `OLLAMA_NUM_PARALLEL=5`, stopped afterwards; the other run used the production service.